

A two-stage sign language generation framework with self-supervised latent representation learning

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ABSTRACT

Sign Language Generation (SLG) aims to convert gloss sequences into sign language sequences. Prevailing methodologies primarily involve designing neural networks that directly predict sign language keypoints from gloss modality. Nevertheless, these sign language results lack detailed information and exhibit limited semantic clarity. To address these shortcomings, we propose a novel two-stage sign language generation framework that indirectly predicts sign language keypoints by leveraging a low-dimensional latent space, capturing high-level semantic information. In the first stage, a Sign Language AutoEncoder (SLAE) employs self-supervised representation learning to abstract sign language keypoints into a low-dimensional representation within the hidden space. In the second stage, we propose a GlossMapper that exploits features from gloss inputs to predict the corresponding low-dimensional hidden space sign language representations. Furthermore, we introduce a Physical Perception loss to enforce constraints on the physical perceptual properties of the generated results. Our experimental results indicate improved fluency in visual effects and clarity in sign language details. On the PHOENIX14T and How2Sign dataset, our method achieves superior performance across multiple evaluation metrics. For example, on the PHOENIX14T dataset, the baseline model achieves a *BLEU* – 2 score of 5.16, whereas our method reaches 6.12. On the How2Sign dataset, the baseline achieves 3.85, while our method attains 5.34. The source code is available at: [repository link](https://github.com/gwfeng1/SLG).

1. Introduction

Sign language is the main mode of communication for the deaf community, and gloss-driven sign language generation research has emerged to improve communication between the normal population and the deaf community [1]. Current research focuses on learning the mapping relationship between gloss modality and sign language directly through neural networks, with particular attention to the accuracy of the predicted hand movements to improve the high intelligibility of the generated sign language. Although all of the above methods have achieved some success, there is still a big flaw in the quality of generation. We have analyzed and summarized the following two problems. (1) **The accuracy of joint locations**; (2) **the impact of expressibility**, such as the inter-frame fluency of sign language key point movements, the realism of the knuckles, and the accuracy of whether they are recognizable by sign language recognition algorithms [2–7]. In previous research, too much attention has been paid to improving the mapping ability of individual

networks. Although the corresponding sign language is successfully predicted from the gloss, there are still bad cases such as low accuracy of hand movements, poor fluency of the generated results, and low semantic information of the generated results. These cannot accurately express the meaning of the sign language. In some other cross-modal prediction tasks [8–11], the transition is made by setting up a reconstructed network based on the codec architecture of the target modal data. Initially, the encoder is employed to get an effective representation of the target modal data. Subsequently, a prediction network is established from the source modality to the hidden space of the target modality, and ultimately, the prediction network and the decoder are combined to get the target modal data.

In various representation learning tasks [12–15], self-supervised mechanisms are widely used to uncover structured representations within data. Initially, by designing pretext tasks such as reconstruction, contrastive learning, or prediction, the model is guided to autonomously learn discriminative latent features from raw inputs. Subsequently, the

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representations obtained through self-supervised learning are transferred to downstream tasks such as classification or generation, demonstrating strong generalization capabilities.

Inspired by the aforementioned work, we propose a two-stage framework for sign language generation that leverages a latent space representation as an intermediate bridge to enable indirect prediction. In the first stage, we employ a self-supervised learning approach to capture the latent space representations of sign language keypoints. These learned self-supervised representations are then transferred to the subsequent sign language generation task. Compared to traditional methods that directly map gloss inputs to sign language keypoints, our indirect prediction strategy more effectively models the complex semantic relationships among keypoints, leading to a significant improvement in prediction accuracy. In addition, we design a Physical Perception loss to ensure the smoothness and accuracy of the generated results in terms of both inter-frame smoothness and keypoint multi-region accuracy. Experiments demonstrate that cross-modal prediction by predicting hidden space representations can lead to more accurate prediction results, while the new loss function makes our results more semantically accurate as well as precise. The contributions of this paper can be summarized as:

- We propose a *SignLanguageAutoEncoder* which capture the latent space representations of sign language keypoints that encode high-level semantic information and a *GlossMapper* by using features of the gloss inputs to predict latent space representations of the corresponding sign language, which aim to improve the accuracy of hand movements and the semantic accuracy of the generated results.
- We propose a Physical Perception loss, which ensures the smoothness and accuracy of the generated results by taking into account both inter-frame smoothness and keypoint multi-region components. The inter-frame loss guarantees smooth transitions between adjacent frames, while the multi-region loss constrains the keypoints of the trunk, left hand, and right hand, enhancing the spatial consistency.
- Multiple experimental results demonstrate that the proposed method significantly improves the fluency and semantic accuracy of sign language generation sequences on the PHOENIX14 and How2Sign datasets. In terms of visual performance, the generated gesture sequences exhibit smooth transitions and coherent motion, effectively mitigating abrupt gesture changes and discontinuities.

2. Related works

The study of sign language generation is the process of transforming sign language vocabulary into gesture movements. However, the temporal relationship between text sequences and sign language sequences is not a sequential correspondence. Consequently, how to effectively learn the mapping relationship between these two types of sequences have emerged as a pivotal challenge within the field. In earlier studies, the mapping between gloss and sign language has been learnt mainly through grammar-based libraries and neural machine translation [16,17]. However, the human gestures generated by these methods are not related to each other and the action representations are rigid. In recent years, Transformer [18–21] and Diffusion [22–25] networks have achieved some results in the field of sign language generation [26,27], such as generating photo realistic sign language videos directly from spoken language using transformer architecture and keypoint-based loss functions [28], or generating continuous sign pose sequences solely from speech segments in an end-to-end manner, leveraging multi-tasking transformers and cross-modal discriminators [29]. However, the generated sign language motion sequences often lack sufficient temporal and spatial contextual information, resulting in poor fluidity and naturalness in the generated sign language videos.

2.1. Transformer architectures

Zelinka et al. simplified the Transformer architecture to convert gloss into sign language sequences [30]. Saunders et al. designed a sign lan-

guage generation network that is able to output sign language sequences of different lengths and used Transformer to learn the mapping relationship between gloss and sign language [31]. Furthermore, Saunders et al. combined Transformer with the GAN network to enhance the accuracy of the generated results [32]. Additionally, they proposed a frame selection network for selecting operations on possible generated sequences and added an additional library of relational mappings to enrich the information of input data, resulting in smoother sign language sequences [33]. However, the sign language generated by the above methods is deficient in terms of the precision of the finger skeletal points and the conspicuousness of the semantic information.

2.2. Diffusion models

Given the extensive application of Diffusion in the field of sign language recognition [9], it is now also being applied in the field of generation. Xie et al. utilized a variational autoencoder (VAE) [34] to model the latent space of sign language sequences [35]. Then they employed the diffusion model to simulate the relationship between gloss and hidden space features, effectively addressing the challenge of constructing mapping networks. This led to the generation of higher quality and more natural sign language results. Conversely, Fang et al. employed the Transformer model for predicting sign language sequences and employed the diffusion model to learn real-life picture features, enabling the style transformation of sign language sequences [36]. Nevertheless, the results indicate that the diffusion-based method exhibits some degree of body jitter during the generation process.

2.3. Self-supervised representation learning

Self-supervised representation learning methods have demonstrated significant improvements in the generalization ability and representational efficiency of sign language generation and recognition models under low-resource conditions. These methods automatically extract universal semantic and motion patterns from unlabeled video data. Zhao et al. proposed a self-supervised contrastive learning framework incorporating spatiotemporal consistency constraints [37], which notably enhances the model's capability in temporal modeling of dynamic sign gestures. Xie et al. introduced the G2P-DDM model [38], which employs a discrete diffusion mechanism to generate corresponding sign pose sequences from gloss annotations, further validating the effectiveness of structured representations in sign language generation tasks. Furthermore, Wong et al. presented the SignRep framework [39], which integrates a skeleton-guided reconstruction task during self-supervised pre-training, thereby improving both the semantic consistency and temporal modeling capacity of the learned representations.

In conclusion, the above methods ignore the motion details of the sign language skeletal points in the generation process, resulting in problems such as poor details of the skeletal point positions and unobvious semantic information in the generation results. Therefore, in our overall framework, we designed a sign language encoder to capture the high-level semantic information of sign language. In addition, we introduced a gloss-based predictor to enhance the semantic understanding of the input information, and proposed a Physical Perception loss to strengthen the constraints on the physical perceptual properties of the results.

Compared with Transformer-based approaches that directly perform cross-modal prediction, our model generates sign language keypoints indirectly through a latent space. In contrast to Diffusion-based methods, we introduce a Physical Perception loss in the decoding stage, which effectively reduces body jitter and enhances spatial consistency. Moreover, unlike most self-supervised methods that focus only on representation learning, our approach enforces physical constraints during generation to further strengthen the consistency between semantics and motion.

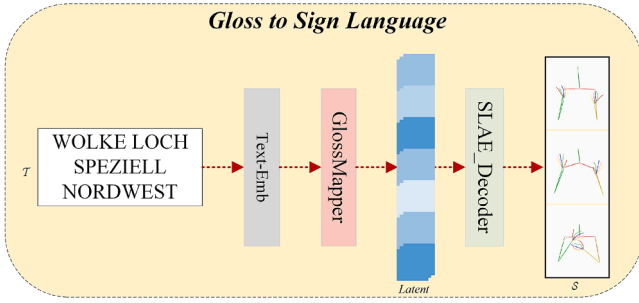


Fig. 1. An overview of the second stage of our method. This method utilizes the sign language latent space predicted by the GlossMapper to guide the generation of the sign language sequence through the SLAE decoder.

3. Methodology

The aim this method is to generate smoother and more accurate sign language sequences from gloss. In the first stage, SLAE captures the latent space representation of sign language keypoints. As shown in Fig. 1, in the second stage, GlossMapper uses gloss to predict the latent space, which is then decoded by SLAE into sign language sequences.

In the following sections, we first detail the format of all the data required for this method (Section 3.1). Subsequently, we describe the SLAE designed in this paper and delve into the design principles of the model's encoder-decoder (Section 3.2). Next, we introduce GlossMapper and elaborate on the method for processing gloss data, as well as the design principles for Physical Perception loss. Finally, we describe the training strategy of our method (Section 3.3).

3.1. Preliminaries

Given a set of input data $(S, \mathcal{T})$, where S represents the sign language sequence $S = (s_1, s_2, \dots, s_N) \in R^{N \times K}$ used as input to the SLAE network. N represents the total number of frames in the sign language sequence. $\mathcal{T}$ denotes the gloss sequence $\mathcal{T} = (t_1, t_2, \dots, t_M)$ used as input to the GlossMapper network. M denotes the number of words in each gloss sequence.

3.2. Sign language autoencoder

In this section we introduce a sign language keypoint reconstruction model for learning the implicit spatial representations of sign language joint points, which consists of an Encoder that abstracts sign language skeletal keypoints into a set of implicit representations and a Decoder that reconstructs sign language skeletal keypoints from them.

The SLAE predicts the latent space representation of sign language keypoints through a self-supervised representation learning approach. By utilizing the encoding-decoding process of an autoencoder architecture, the model effectively learns the deep semantic relationships and temporal dependencies between sign language keypoints. The decoder of SLAE acts as a guide, driving the GlossMapper predictor to accurately predict the latent space representation of sign language.

Encoder. We apply a Embedding layer $SL_Emb_{(512)}$ to embed the sign language key point sequence to a high-dimensional space, aiming to extract the key information effectively:

$$F_S = SL_Emb_{(512)}(S) \quad (1)$$

Subsequently, we devise a Former network, which incorporates a self-attention mechanism, to learn the inter-frame correlation of sign language sequence features. This helps to capture the continuity characteristics of sign language expressions in the time dimension. The dynamic information in the sign language information is then captured, resulting in a hidden space representation $Latent_{SLAE}$, where $Q, K, V =$

F_S :

$$Scores(Q, K) = Softmax\left(\frac{Q}{\sqrt{Head_Size}} \otimes K\right) \quad (2)$$

$$Attention(Q, K, V) = Softmax(Scores(Q, K) \otimes V) \quad (3)$$

$$Latent_{SLAE} = Linear(Attention(Q, K, V)) \quad (4)$$

Decoder. Compared with models such as transformer, BiLSTM can better captures long-term dependencies and avoids a lot of matrix operations. Moreover, it can capture past and future contextual information while achieving efficient temporal reconstruction with lower computational cost. Therefore our decoder receives the hidden space features as input and uses BiLSTM network to obtain the output features. Immediately after that, the feature is reconstructed through a linear layer to obtain the corresponding sequence of sign language keypoints $\tilde{S}_R$:

$$\tilde{S}_R = Linear(\overline{BiLSTM}(Latent_{SLAE})) \quad (5)$$

The Encoder and decoder can be trained end-to-end by using MSE metrics for input and output data as follow:

$$\mathcal{L}_{SLAE} = \frac{\sum_1^N \frac{\sum_1^{K-1} (x_i - x_{i+1})^2}{K}}{N} \quad (6)$$

3.3. GlossMapper

In this section, we introduce GlossMapper for predicting the hidden space of sign language through Gloss features, as Fig. 2 shows.

Gloss Encoding. When given gloss sequences of indeterminate length, it is not possible to perform uniform operations due to the inconsistent length of each gloss sequence. To solve this problem, we develop a maximum input gloss length and padded shorter sequences for subsequent processing. We construct a vocabulary containing all gloss words for the embedding layer $Gloss_Emb_{(512)}$ to convert discrete gloss words into continuous features. Then, we use a multi-layer convolutional architecture to encode local, structural, and semantic features of the input gloss. Each convolutional block extracts local contextual information of the input gloss at different scales, while stacking multiple layers gradually expands the sensory field to capture a larger range of spatial structural information. This allows the model to capture details in the lower layers and broader semantic information in the deeper layers. Through the convolutional blocks, we are able to gradually extract a high-level feature $F_{\mathcal{T}} = (f_{t_1}, f_{t_2}, \dots, f_{t_M})$ of the gloss sequence:

$$F_{\mathcal{T}} = Conv_{1D(5 \times 1)}^3(Gloss_Emb_{(512)}(\mathcal{T})) \quad (7)$$

Gloss Former. To further enhance the expressive power of the sequential features, we employ a former to capture the contextual information of the sequence features, denoted as $Latent_{\mathcal{T}}$:

$$Scores(Q_{\mathcal{T}}, K_{\mathcal{T}}) = Softmax\left(\frac{Q_{\mathcal{T}} \otimes K_{\mathcal{T}}}{\sqrt{Head_Size}}\right) \quad (8)$$

$$Attention(Q_{\mathcal{T}}, K_{\mathcal{T}}, V_{\mathcal{T}}) = Softmax(Scores(Q_{\mathcal{T}}, K_{\mathcal{T}}) \otimes V_{\mathcal{T}}) \quad (9)$$

$$Latent_{\mathcal{T}} = Attention(Q_{\mathcal{T}}, K_{\mathcal{T}}, V_{\mathcal{T}}) \quad (10)$$

where $Q_{\mathcal{T}}, K_{\mathcal{T}}, V_{\mathcal{T}} = F_{\mathcal{T}}$. Then feed $Latent_{\mathcal{T}}$ into the co-trained Decoder to obtain the final predicted sign language keypoint features $\tilde{S}_{\mathcal{T}}$:

$$SLAE_Decoder = Linear(\overline{BiLSTM}) \quad (11)$$

$$\tilde{S}_{\mathcal{T}} = SLAE_Decoder(Latent_{\mathcal{T}}) \quad (12)$$

3.4. Physical Perception loss

In order to optimise the coherence between every two frames of the generated sign language keypoint sequence and ensure the smoothness of the generated results, we design a Physical Perception loss, which is used to constrain the physical perception attributes of the generated

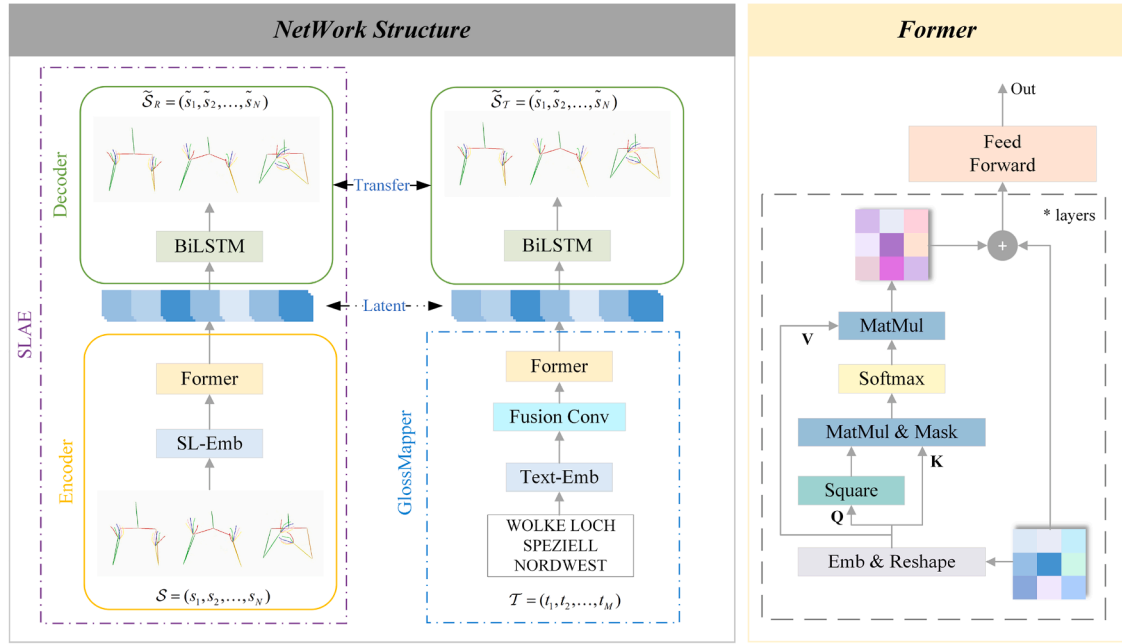


Fig. 2. The structure of proposed method. SLAE: A sign language reconstruction network built with a auto-encoder for predicting sign language hidden space. GlossMapper: Gloss based hidden space prediction network for sign language. In addition, we design a Forner network for enhancing the temporal information in sign language features or gloss features.

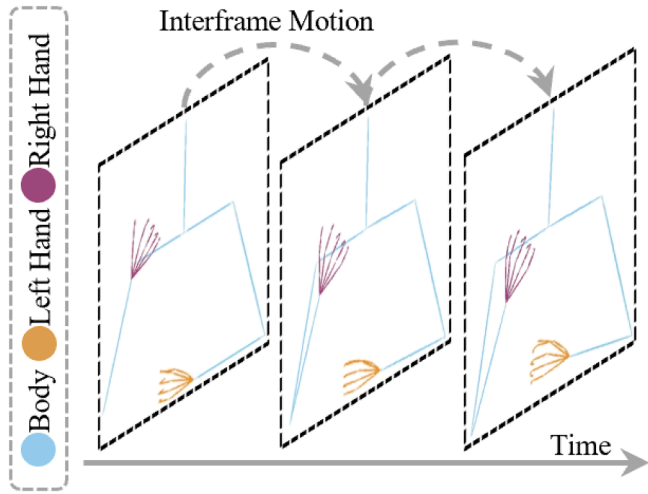


Fig. 3. Visualization of the Physical Perception loss setting. This function constrains the physical perceptual properties of the generated results in two ways: $\mathcal{L}_{\text{inter-frame}}$ and $\mathcal{L}_{\text{multi-region}}$. The joints are precisely constrained by dividing them into three parts: the body, the left hand, and the right hand.

results, including the generation of the results in terms of the visual coherence, smoothness and the accuracy of the results:

$$\mathcal{L}_{\text{PhysicalPerception}} = \mathcal{L}_{\text{inter-frame}} + \mathcal{L}_{\text{multi-region}} \quad (13)$$

This function consists of the inter-frame loss $\mathcal{L}_{\text{inter-frame}}$:

$$Pre_{frame} = Pre([1 : n] - [0 : m - 1]) \quad (14)$$

$$GT_{frame} = GT([1 : n] - [0 : m - 1]) \quad (15)$$

$$\mathcal{L}_{\text{inter-frame}} = \text{MSE}(Pre_{frame}, GT_{frame}) \quad (16)$$

which measures the difference between the prediction and the reality by judging the displacement of joints between the neighboring frames to ensure the coherence and fluency of the results, such as Fig. 3 shows. In this equation, Pre and GT are the predicted and real sign language

sequences, respectively. And a multi-region loss function $\mathcal{L}_{\text{multi-region}}$ for constraining each part of the body separately:

$$\mathcal{L}_{\text{multi-region}} = \alpha * \mathcal{L}_{\text{trunk}} + \beta * \mathcal{L}_{\text{left-hand}} + \gamma * \mathcal{L}_{\text{right-hand}}, \quad (17)$$

since the movement amplitude of the body part is smaller than the hands during the sign language movement process, we set the weights of the three parts as $\alpha = 0.2$, $\beta = 0.4$, $\gamma = 0.4$, respectively, according to the actual experience, so as to realize the accuracy control of the generated results.

4. Experiment

4.1. Dataset

We use RWTH-PHOENIX-Weather 2014 T [40] and How2Sign [41] dataset in our work. PHOENIX14T provides RGB videos and their corresponding text annotations [42]. It comprises 7096 training samples, 519 validation samples, and 642 test samples. Due to significant variations in the number of frames across samples in the dataset, we mainly select data ranging between 50 and 100 frames for both training and evaluation purposes in order to allow the network to learn better. The How2Sign training dataset consists of 31,048 video segments, with 1739 segments in the development set and 2343 segments in the test set.

4.2. Implementation details

In this study, we utilize a total of 50 joints obtained from the human body, which were extracted using the OpenPose framework [43]. However, some hand joints could not be successfully extracted due to the limited resolution of the raw data and occlusion problems in the data. To address this issue, we employ inverse kinematics to supplement the missing key points and obtain a 3D coordinate representation of all 50 joints [30]. In our method, the transformer module has a d_{model} value of 512, and it consists of a total of 2 layers and 8 heads. During the training process, our method uses NVIDIA 3090 GPU for training. The joint training procedure of our method is shown in Algorithm 1.

Algorithm 1 Co-training with SLAE and GlossMapper.

Input: sign language video S_V , gloss text $\mathcal{T}_G$
Output: sign language prediction sequence $\tilde{S}_T$
Preprocessing
1. Extracting sign language keypoints from S_V using OpenPose:
 $S_{point} = OpenPose(S_V)$
2. Supplementing missing S_{point} using Inverse Kinematics (IK) to obtain S :
 $S(s_1, s_2, \dots, s_i) = IK(S_{key})$
3. Convert gloss text to vectors using a glossary:
 $\mathcal{T}(t_1, t_2, \dots, t_i) = Glossary(\mathcal{T}_G)$
gr1 Begin
1: **for** s_i in S **do**
2: // Obtaining latent space representation
3: $Latent_{s_i} = SLAE_{Encoder}(s_i)$
4: // Reconstructing sign language sequences
5: $\tilde{s}_{iR} = SLAE_{Decoder}(Latent_{s_i})$
6: // Compute loss
7: $\mathcal{L}_{SLAE} = f_{MSE}(\tilde{s}_{iR}, s_i)$
8: **for** t_i in $\mathcal{T}$ **do**
9: // Obtain predicted hidden spatial representations
10: $semantic_i = GlossMapper(t_i)$
11: // Co-training with SLAE's Decoder
12: $\tilde{s}_{iT} = SLAE_{Decoder}(semantic_i)$
13: // Compute loss
14: $\mathcal{L}_{SLGM} = f_{MSE}(\tilde{s}_{iT}, s_i)$
15: $\mathcal{L}_{PhysicalPerception} = \mathcal{L}_{inter-frame} + \mathcal{L}_{multi-region}$
16: $\mathcal{L}_{GlossMapper} = \mathcal{L}_{SLAE} + \mathcal{L}_{SLGM} + \mathcal{L}_{PhysicalPerception}$
17: **return** $\tilde{S}_T$

4.3. Evaluation metrics

Similar to previous studies, we use *BLEU* as an assessment metric [44]. $BLEU_n$ represents the weighted average precision of the most significant $n - gram$ matches:

$$P_n = \frac{\sum_{C \in generate\ n-gram \in C} Count_{clip}(n - gram)}{\sum_{C' \in generate\ n-gram' \in C'} Count_{clip}(n - gram')} \quad (18)$$

$$BLEU_N = BP * exp(\sum_{n=1}^N \omega_N * \log(P_n)) \quad (19)$$

Table 1
Comparison with existing methods.

PHOENIX14T(German)						
Method	Publisher	Year	BLEU1↑	BLEU2↑	BLEU3↑	BLEU4↑ ROUGE↑
GroundTruth			24.25	15.7	11.44	9.06 23.08
PTR [31]	ECCV	2020	11.65	5.16	3.33	2.26 11.04
PTR-AT [32]	arXiv	2020	11.47	5.27	3.49	2.38 10.91
SIGNGAN [27]	CVPR	2022	10.88	5.69	4.09	2.91 10.33
ASL-SLP [36]	arXiv	2023	11.20	5.60	4.02	2.95 10.72
PoseVQ-Diffusion [38]	AAAI	2024	10.79	5.43	3.85	2.77 10.19
Ours			11.84	6.12	4.37	3.16 11.69
How2Sign(English)						
Method	Publisher	Year	BLEU1↑	BLEU2↑	BLEU3↑	BLEU4↑ ROUGE↑
GroundTruth			16.15	7.66	3.95	2.13 11.40
PTR [31]	ECCV	2020	9.87	3.85	1.82	0.76 9.95
PTR-AT [32]	arXiv	2020	9.91	3.92	1.91	0.79 9.98
SIGNGAN [27]	CVPR	2022	9.93	3.92	1.91	0.80 9.96
ASL-SLP [36]	arXiv	2023	10.06	3.96	1.93	0.80 10.10
PoseVQ-Diffusion [38]	AAAI	2024	6.70	2.81	1.54	0.88 7.26
Ours			10.27	5.34	2.82	1.15 8.09

where $Count_{clip}$ determines the maximal similarity length between the reference and generated sequences, and P_n quantifies the precision of individual $n - gram$. As the value of n increases, the metric's focus shifts accordingly, emphasizing different aspects of the data. When n has a lower value, the focus is on the accuracy of the generated output, while for higher n values, the assessment shifts towards evaluating the semantic coherence of the generated output. In our evaluation, we employ $BLEU - 1$, $BLEU - 2$, $BLEU - 3$, and $BLEU - 4$ metrics to comprehensively evaluate and measure the quality of our obtained results.

In addition, we adopt *ROUGE* as another evaluation metric to focus on the shortcomings of *BLEU* in terms of the completeness of the generated sequences. *ROUGE* is an evaluation method oriented to $n - gram$ recall, which evaluates the quality of the generated results by counting the number of overlapping basic units between the true values and the generated sequences, and improves the stability and robustness of the evaluation system. $n - gram$ is to fetch data in a text sequence in bytes, using a sliding window of size N for the fetch operation and calling each byte fragment a *gram*.

The calculation can be expressed by:

$$ROUGE - N = \frac{\sum_{S \in GT} \sum_{gram_n \in S} Count_{match}(gram_n)}{\sum_{S \in GT} \sum_{gram_n \in S} Count(gram_n)}, \quad (20)$$

where the denominator represents the number of $n - gram$ in the true value and the numerator represents the number of $n - gram$ overlapping between the true value and the generated result.

4.4. Comparison to existing methods

Table 1 shows the result metrics for our method and existing methods on the PHOENIX14T and How2Sign dataset. For the PTR model [31], we replicated the results for comparative analysis. In addition, due to the lack of open source code and the inconsistency of the baseline results in each paper [27,32,36,38], we endeavored to replicate the code of other models according to the methodology described in the paper, and then obtained the metrics results. Additionally, the presence of “↑” in the table indicates that higher values of the indicator correspond to better performance.

As shown Table 1, our proposed method significantly outperforms existing approaches across all evaluation metrics on the PHOENIX14T dataset, demonstrating a marked improvement in the semantic accuracy of the generated results. On the How2Sign dataset, although the *ROUGE* scores are slightly lower than existing methods, our approach achieves superior performance on $BLEU - 1$ to $BLEU - 4$ scores, indicating stronger capabilities in keyword and phrase matching. This discrepancy primarily stems from the flexibility and diversity of English expressions, which makes *ROUGE* more sensitive to surface-level overlaps. In contrast, our model focuses on the semantic reconstruction and refined expression of sign language sequences, thereby exhibiting higher accuracy in conveying key information.

In addition, as shown in Fig. 4, we list some visualisation results of the comparison experiments on the PHOENIX14T dataset. The blue dashed box in the figure illustrates that the existing methods are unable to distinguish individual fingers or to accurately represent the objective laws of skeletal points in sign language. In contrast, our method is significantly more accurate in capturing finger movements. In Fig. 5, real-person images are presented along with the corresponding visualizations of the predicted sign language keypoints. Furthermore, additional comparison videos are provided in the supplementary material.

4.5. Ablation study

Effectiveness of Our Method. In addition, we conduct ablation experiments on our method utilizing the PHOENIX14T and How2Sign dataset and analysed its effectiveness based on the experimental results, as shown in Table 2. The “PPlOSS” denotes the physical perceptual loss

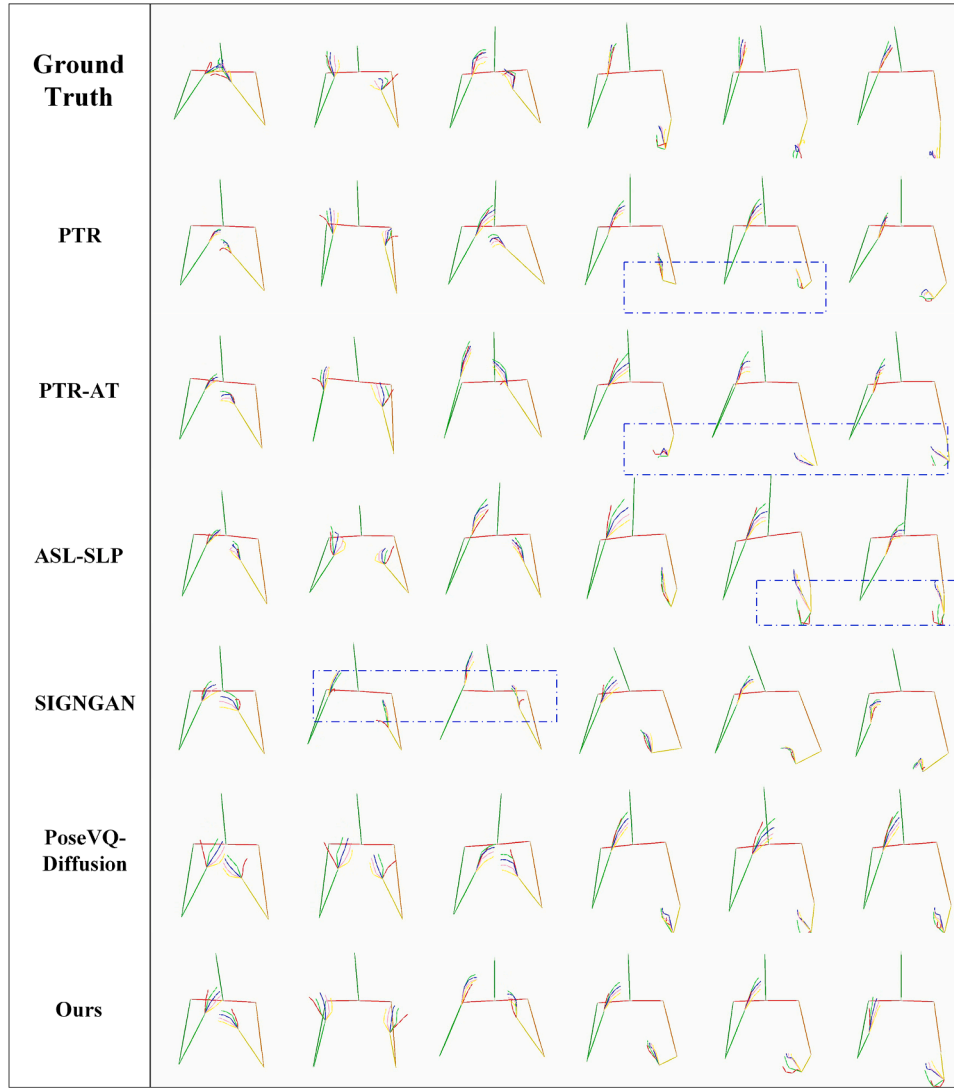


Fig. 4. Sign language production result. The first row represents the real sign language sequence, the last row shows the result generated by our method, and the remaining rows show the results of existing methods. Video result comparisons are presented in the supplementary material.

Table 2

Ablation study on component contribution. '✓' indicates the module is included, and '×' means it is not.

PHOENIX14T(German)						
PPloss	Future	BLEU1↑	BLEU2↑	BLEU3↑	BLEU4↑	ROUGE↑
GroundTruth		24.25	15.70	11.44	9.06	23.08
SLAE (Reconstruction)		22.85	14.89	11.08	8.92	22.36
×	×	10.71	5.36	3.77	2.69	10.25
✓	×	10.82	5.40	3.87	2.82	10.46
✓	✓	11.84	6.12	4.37	3.16	11.69
How2Sign(English)						
PPloss	Future	BLEU1↑	BLEU2↑	BLEU3↑	BLEU4↑	ROUGE↑
GroundTruth	16.15	7.66	3.95	2.13	1.40	
SLAE (Reconstruction)		14.23	5.89	2.92	1.93	10.93
×	×	8.29	3.63	1.82	0.79	7.58
✓	×	10.08	5.11	2.68	1.11	8.01
✓	✓	10.27	5.34	2.82	1.15	8.09

Table 3

Hyperparameter settings.

Hyperparameter	Value
batch_size	128
learning_rate	0.001
learning_rate_min	0.00002
validation_freq	10,000

Table 4

Continuation of Table 3.

Hyperparameters	Value
future_prediction	10
future_prediction_from	0
random_seed	27
trg_size	150

proposed in this study, “Future” denotes the expansion of the sign language sequence data in the temporal sequence.

As discerned from the tabulated results, our SLAE method achieves no difference from the true value on the sign language reconstruction task. Furthermore, the experimental results demonstrate that adding the Physical Perception loss significantly improves the performance of the

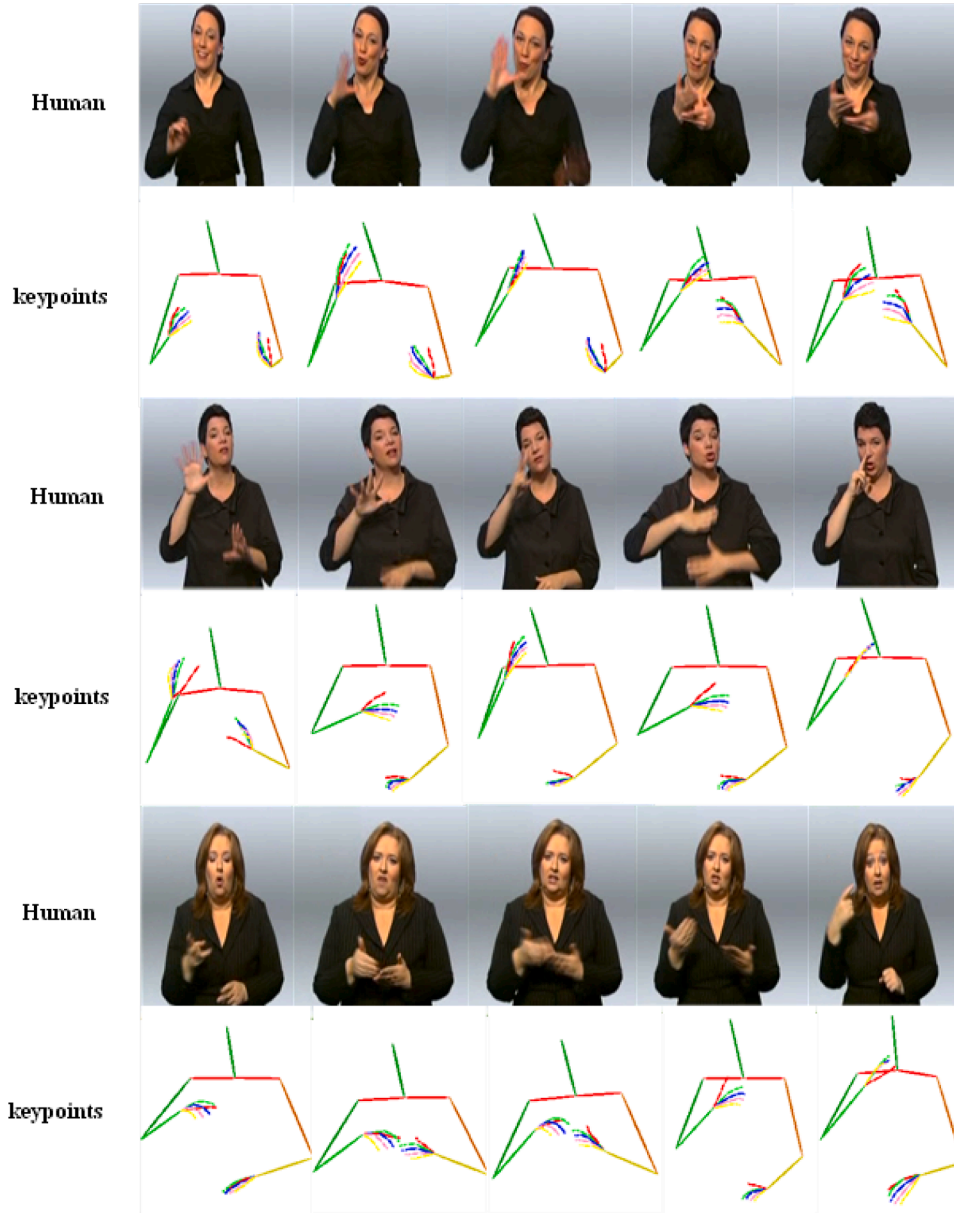


Fig. 5. Sign language keypoint prediction results. The first row of each group shows the real-person images, while the second row presents the predicted keypoints. These results demonstrate the model's ability to accurately capture hand and upper limb movements during sign language performance.

basic network. This improves the accuracy of the generated results and proves the effectiveness of each module of our method.

Effectiveness of SLAE Latent Space. The quality of the predicted latent space representations is crucial for the final quality of our indirect sign language generation method. To visualize the distribution of the latent space data, we use t-SNE to reduce it to 2D features and plot the resulting data points. In a scatter plot, data from the same sample are clustered together, and different samples are clearly distinguished. The results are shown in Fig. 6. In both part A and B, our predicted latent space representations closely match SLAE's latent space in terms of aggregation. Furthermore, the part C and D show that in our predicted latent space, each sign language sequence exhibits a concentrated and continuous distribution of dots, indicating strong temporal continuity. Additionally, the latent space of each data point is also distinct from the other data points.

Hyperparameter Settings. To ensure the validity of the hyperparameters set in the experiments, we also performed ablation experiments for the set of the hyperparameters. In Tables 3 and 4, the default settings of

selected hyperparameters are presented. In Table 5, the "Value" column represents the value of the hyperparameter, which is $[Embeddingdim, \alpha, \beta, \gamma]$. Since we need to apply Transformer in our method, we set the values of 64, 128, etc. for Embedding dim to test. In addition, α , β , and γ are the weight coefficients for the upper body, left hand, and right hand, respectively, within the Physical Perception loss component. To systematically examine their influence, we designed four sets of weight combinations with different motivations: [1, 1, 1] serves as a baseline where all parts are treated equally, used to verify whether assigning higher weight to the hands is necessary; [1, 2, 2] emphasizes the hands by setting their weights to twice that of the upper body; [0.4, 0.8, 0.8] preserves this ratio but lowers the total sum of the weights to assess the effect of scaling; and [0.2, 0.4, 0.4] also maintains the ratio while normalizing the total weight to 1, making it consistent with common practices in loss function design.

From the experimental results, it can be seen that the best results are obtained at hyperparameters of [512, 0.2, 0.4, 0.4]. Experimental results on the PHOENIX14T and How2Sign datasets show that the model

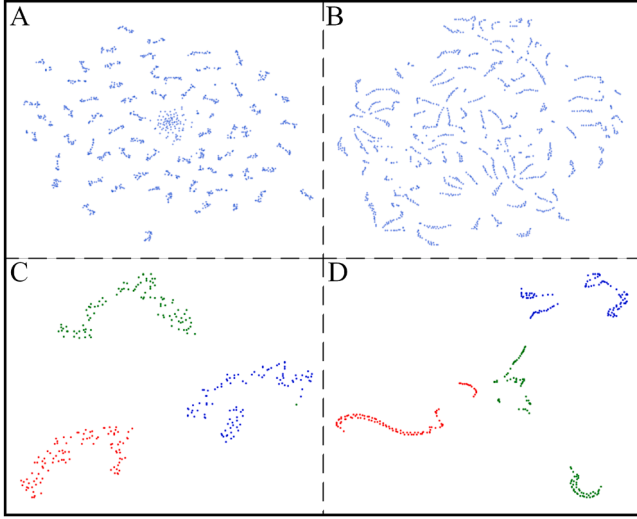


Fig. 6. Visualisation of the hidden space variables for gloss data prediction and for sign language. In the figure, A and B represent the distributions of text prediction and sign language, respectively, with a data size of 64. C and D represent the distributions of text prediction and sign language, respectively, and we randomly select samples from the dataset with a data size of 3. Each color represents a sign language sequence.

Table 5
Ablation study on hyperparameter setting.

PHOENIX14T(German)						
	Value	BLEU1↑	BLEU2↑	BLEU3↑	BLEU4↑	ROUGE↑
Embedding dim	[64, 0.2, 0.4, 0.4]	10.83	5.53	3.89	2.75	10.29
	[128, 0.2, 0.4, 0.4]	10.91	5.50	3.91	2.84	10.37
	[256, 0.2, 0.4, 0.4]	11.43	5.95	4.23	3.01	11.21
	[512, 0.2, 0.4, 0.4]	11.84	6.12	4.37	3.16	11.69
PP loss weight	[512, 1, 1, 1]	11.16	5.58	3.96	2.84	10.25
	[512, 1, 2, 2]	11.02	5.62	3.98	2.82	10.35
	[512, 0.4, 0.8, 0.8]	11.21	5.71	4.04	2.88	10.43
	[512, 0.2, 0.4, 0.4]	11.84	6.12	4.37	3.16	11.69
How2Sign(English)						
	Value	BLEU1↑	BLEU2↑	BLEU3↑	BLEU4↑	ROUGE↑
Embedding dim	[64, 0.2, 0.4, 0.4]	8.39	3.61	1.75	0.78	7.48
	[128, 0.2, 0.4, 0.4]	9.92	5.04	2.67	1.11	7.95
	[256, 0.2, 0.4, 0.4]	9.74	3.70	1.68	0.72	9.26
	[512, 0.2, 0.4, 0.4]	10.27	5.34	2.82	1.15	8.09
PP loss weight	[512, 1, 1, 1]	9.81	4.90	2.54	1.08	7.85
	[512, 1, 2, 2]	10.11	5.20	2.68	1.10	8.06
	[512, 0.4, 0.8, 0.8]	8.32	3.65	1.82	0.79	7.63
	[512, 0.2, 0.4, 0.4]	10.27	5.34	2.82	1.15	8.09

exhibits high sensitivity to key hyperparameter. In particular, as the embedding dimension increases, both *BLEU* and *ROUGE* scores show a steady upward trend overall, indicating that greater representational capacity helps capture richer semantic information.

Moreover, the weight distribution of different body parts in the Physical Perception loss also has a significant impact on model performance. Experiments show that the model achieves the best performance on both datasets when the weight of the hand region is set to twice that of the upper body, and the weights of all parts are normalized to sum to 1. In contrast, setting overly large weights leads to dispersed model attention, negatively affecting the optimization of key targets. These results highlight the importance of fine-tuning the weights of different body parts in the loss function, which is especially critical in sign language generation tasks dominated by hand movements.

Table 6
User study.

Method	Publisher	Year	GC↑	GN↑	MF↑	SA↑	FinalScore↑
PTR [31]	ECCV	2020	3.80	2.90	3.00	2.80	3.17
PTR-AT [32]	arXiv	2020	2.60	2.90	2.30	3.00	2.71
SIGNGAN [27]	CVPR	2022	3.20	2.40	3.10	2.10	2.72
ASL-SLP [36]	arXiv	2023	3.20	4.00	3.40	3.10	3.46
PoseVQ-Diffusion [38]	AAAI	2024	2.50	1.60	1.50	3.00	2.13
Ours			4.60	3.80	4.90	4.20	4.34

4.6. User study

We conducted a user study by inviting 10 non-sign language users to evaluate the performance of our model on multiple metrics (rated 1-5) and compared it with other methods. The details are shown in Table 6. The subjective evaluation of the generated results was performed based on four key dimensions: *Gesture Clarity (GC)*, *Gesture Naturalness (GN)*, *Motion Fluency (MF)*, and *Spatial Accuracy (SA)*. Since the evaluators were non-sign language users, we chose not to conduct the study at the semantic level. The final score was calculated using the following formula:

$$FinalScore = \alpha_{GC} * GC + \alpha_{GN} * GN + \alpha_{MF} * MF + \alpha_{SA} * SA \quad (21)$$

In sign language generation, the most important factors are the naturalness and clarity of the gestures, as these directly affect the transmission and understanding of information. Therefore, we assigned the following weights to each dimension: $\alpha_{GC} = 0.3$, $\alpha_{GN} = 0.3$, $\alpha_{MF} = 0.2$, $\alpha_{SA} = 0.2$.

According to the results in Table 6, our method achieves the highest scores in three out of four user study metrics, and the final score is significantly higher than all compared methods. In particular, the Motion Fluency score is substantially higher than that of existing approaches, clearly demonstrating the advantage of our method in generating smooth and coherent gesture sequences. Moreover, it is necessary to expand the study in the future and involve both sign language users and deaf participants to further enhance the authority and reliability of the evaluation.

5. Conclusion

In this paper, we propose a method that utilises the hidden space to generate sign language, which improves the fluency as well as the semantic accuracy of the keypoints of sign language on the PHOENIX14T and How2Sign dataset. Comparative experiments demonstrate that our method outperforms existing approaches in both accuracy and fluency of sign language generation, while ablation studies further verify the effectiveness of the proposed Physical Perception loss, which notably enhances the spatial consistency and temporal coherence of the generated gestures. However, the current method still has some limitations. (1) Our proposed method is not characterized as an end-to-end architecture, which may lead to efficiency problems; (2) the method is only superficial in predicting the latent space, particularly when the multi-modal data fusion approach is not thoroughly investigated. To address these issues in future work, we plan to extend the current two-stage framework into an end-to-end architecture by employing a shared encoder and joint optimization objectives. Furthermore, we plan to integrate speech as an additional modality in latent space prediction and leverage cross-modal alignment and attention mechanisms to achieve deeper fusion of speech and text, enhancing the semantic accuracy of the generated results. Additionally, we aim to collaborate with the deaf community to gain deeper insights and feedback, while also focusing on tasks related to generating realistic sign language gestures.

CRedit authorship contribution statement

Qiguang Miao: Writing – review & editing, Writing – original draft, Project administration, Funding acquisition; **Guanwen Feng:** Writing – review & editing, Writing – original draft, Project administration, Methodology; **Junwei Jing:** Writing – review & editing, Writing – original draft, Visualization, Conceptualization; **Yilin Zhang:** Formal analysis, Data curation, Conceptualization; **An Liu:** Software, Investigation; **Yunan Li:** Investigation, Formal analysis, Data curation; **Chi-Man Pun:** Validation, Supervision, Conceptualization.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary material

Supplementary material associated with this article can be found, in the online version, at [10.1016/j.knosys.2025.114979](https://doi.org/10.1016/j.knosys.2025.114979)

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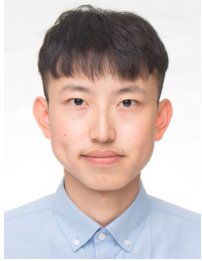
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